1. Find the equation of the tangent line and the equation of the normal (or perpendicular) line to the curve $y = 3\sqrt{x} + 2x - 1$ at the point (1,4).

2. Differentiate the function $f(x) = \frac{6t+1}{6t-1}$.

3. $f(x) = 2t^4 + 3t^3 - 2$, find f"(2)

4. For what value does the curve $f(x) = x^3 + 3x^2 + x + 3$ have a horizontal tangent?

5. $\frac{d}{dx}\left(x^2 \sin(x) \tan(x)\right) =$